provides the required braking force for controlling the speed of vehicle. However, suddenly or unexpectedly when the vehicle is stopped, the vehicle is pulled away from its normal braking due to the road condition, condition of the tread and vehicle load etc. This situation leads to accidents.

To avoid this situation, before the vehicle is started to pull along one side, the antilock braking system provides braking force to stop the wheel by giving and releasing the force intermittently. It is a kind of special type of braking system called as ABS system. In this system a hydraulic control unit and an electronic control unit are considered as important components. ECU receives the signals based on the various programming situations from the wheels (input signals) under various conditions and converts them into input control signals and then sends to the hydraulic control unit.

Hence the system opens the brake fluid valve intermittently and prevent the wheel to lock completely. The block diagram of the ABS can be seen in Figure 6.6 and the effect of ABS on the vehicle during braking can be seen in Figure.

Advantages of A.B.S system

1. It prevents the vehicle pulling from one side during braking.
2. Reduces the stopping distance of the vehicle wheels.
3. Brake lining and the parts of the brake which withstand the vibration could work for long time
4. It is possible to drive and steer the vehicle under control even during risky suituations.

6.7 Hand brake

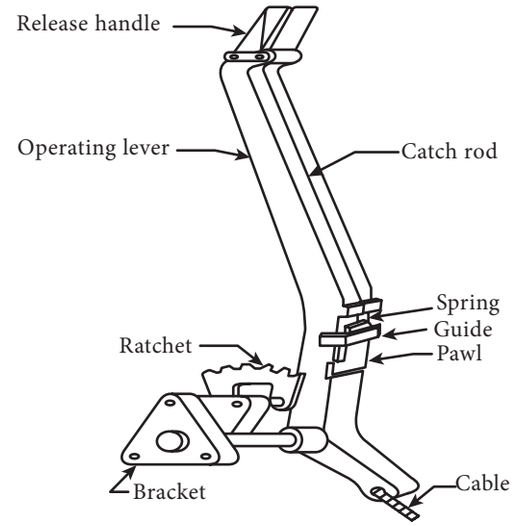


Figure 6.7 Hand Brake

During normal operating conditions, for stopping the vehicle foot brake is commonly used. However, during emergency or dangerous situations when the foot operated brake loses its ability to operate correctly, then hand brake is used to stop the vehicle. Moreover when the vehicle is climbing down on lower roads and when the vehicle is under unbalanced condition of the load, the hand brake is used to avoid movement of vehicle automatically. As the hand brake is use to stop the vehicle and bring it to its balanced condition it is also called as parking brake.

Hand brake is totally separated from the foot brake system. Mostly mechanical type brake is used in the hand brake system. The hand brake is connected with the rear wheels which receive the final drive. The hand operated brake lever is placed on the right side of the driver's seat.

Working principle

Hand brake works based on the ratchet and pawl mechanism. When the

BAE_ENG_Chapter_06.indd 113 12/20/2019 1:04:23 PM